US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250281867
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Hourani; Monzer A. et al.

PURIFICATION DEVICE HAVING HEATED FILTER FOR KILLING BIOLOGICAL SPECIES, INCLUDING COVID-19

Abstract

A purification device and method for treating air flow of an air handling system. A high-efficiency particulate air (HEPA) filter can be disposed across a plenum of the purification device that is configured for passage of the air flow. A heater can be disposed across the plenum and may have a permeable barrier that is configured to allow the air flow to pass therethrough. A metal material of the permeable barrier can be connected in electrical communication to a supplied power. The permeable barrier can have an active surface area configured to interact with a pathogen of the air flow and can be heated by the supplied power to a surface temperature directed to kill the pathogen.

Inventors: **Hourani; Monzer A. (Houston, TX), Ren; Zhifeng (Houston, TX), Yu; Luo (Houston, TX)**

Applicant: **University of Houston System** (Houston, TX)

Family ID: **1000008619565**

Appl. No.: **19/216414**

Filed: **May 22, 2025**

Related U.S. Application Data

parent US continuation 16883977 20200526 parent-grant-document US 12330107 child US 19216414
us-provisional-application US 63018442 20200430
us-provisional-application US 63018448 20200430

Publication Classification

Int. Cl.: **B01D46/42** (20060101); **A61L9/20** (20060101); **B01D39/20** (20060101); **B01D46/00** (20220101); **B01D46/44** (20060101); **B60H3/06** (20060101); **F24F8/10** (20210101);

F24F8/22 (20210101)

U.S. Cl.:

CPC **B01D46/4263** (20130101); **A61L9/20** (20130101); **B01D39/2027** (20130101); **B01D46/0028** (20130101); **B01D46/4245** (20130101); **B01D46/429** (20130101); **B01D46/444** (20130101); **B01D46/448** (20130101); **B60H3/0608** (20130101); **F24F8/10** (20210101); A61L2209/11 (20130101); A61L2209/111 (20130101); A61L2209/12 (20130101); A61L2209/14 (20130101); A61L2209/15 (20130101); B01D2279/50 (20130101); B01D2279/65 (20130101); F24F8/22 (20210101); F24F2221/125 (20130101); F24F2221/34 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 16/883,977, filed May 26, 2020, which claims priority to U.S. Provisional Appl. Nos. 63/018,442 and 63/018,448, both filed Apr. 30, 2020, the entire contents of each of which is incorporated herein by reference. This application relates to co-pending U.S. application Ser. No. 16/883,981, filed on May 26, 2020 (now abandoned), and its continuation U.S. application Ser. No. 18/213,644, filed on Jun. 23, 2023 (co-pending), both entitled “Mobile Purification Device Having Heated Filter for Killing Biological Species, Including COVID-19”, the subject matter of each of which is incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE
[0002] Various infectious pathogens, including bacterium, viruses, and other microorganism can cause disease in humans. The deadly Human SARS-COV-2 strain (COVID-19) pandemic has impacted the human condition at all levels of life as we know it around the globe. The COVID-19 infection is persistently spread by circulating air flow as the primary mechanism for transmission. There are few active strategies to protect the public against COVID-19, and current strategies are widely debated, costly, and inefficient. A passive approach to condition and purify circulating air in all environments is needed to combat aerosolized COVID-19 immediately because current filter and air-purification technologies are not successful at killing the small sized (0.05-0.2 microns) COVID-19 virus.
[0003] Overall, air filtration is used in heating, ventilating, and air conditioning (HVAC) systems to remove dust, pollen, mold, particulates, and the like from the air being moved through a facility by the system. The filters used for the filtration can come in a number of forms and can be configured to filter particles of a given size with a given efficiency.
[0004] For example, high-efficiency particulate air (HEPA) filters are commonly used in cleanrooms, operating rooms, pharmacies, homes, etc. These filters can be made of different types of media, such as fiberglass media, ePTFE media, etc., and may have activated carbon-based material. In general, HEPA filters can filter over 99 percent of particles with a diameter of a given size (e.g., 0.3 microns or larger in size). Even with their efficiency, HEPA filters may not stop pathogens (virions, bacteria, etc.) of very small size.
[0005] Ultraviolet (UV) germicidal lights can stop pathogens, such as bacteria, viruses, and mold. The UV germicidal lights produce ultraviolet radiation, which can then damage the genetic material of the microorganisms. The damage may kill the pathogen or make them unable to reproduce. Extended exposure to the UV radiation can also break down pathogens that have deposited on an irradiated surface.
[0006] One example of an ultraviolet system includes an upper room air ultraviolet germicidal irradiation (UVGI) system. In the UVGI system, the UV germicidal light is installed near the

ceiling in an occupied room. Air circulated by convection near the ceiling in the upper portion of the space is then irradiated within an active field of the UV germicidal light. UVGI systems can also be installed in the ducts of HVAC systems and can irradiate the small airborne particles containing microorganisms as the air flows through the ducts.

[0007] Although existing systems for filtration and germicidal irradiation can be effective in treating air to remove particulates and damage pathogens, there is a continuing need to purify air in populated environs, such as facilities, homes, workspaces, hospitals, nursing homes, sporting venues, and the like, to reduce the spread of pathogens, such as bacteria, viruses, and molds, even more.

[0008] In particular, the 2019 novel coronavirus disease (COVID-19) is a new virus of global health significance caused by infection of severe acute respiratory syndrome coronavirus 2 (SARS-COV-2). COVID-19 is thought to spread from person to person in close contact through respiratory droplets. Studies show the virus can survive for hours at a time and can be persistently carried by airflow. For this reason, it is believed that a stationary 6-feet separation is ineffective in a situation where people spend a long time together in a room because infection can simply be carried by the airflow.

[0009] For example, COVID-19 (Sars-COV-2) may survive in droplets for up to three hours after being coughed in the air, and convection in the air is thought to be the primary mechanism for the spread of the infection. Accordingly, droplet-spray and convection can drive direct airborne infection, and social distancing can be ineffective for enclosed environments were people spend a long time together.

[0010] As there is no current cure for COVID-19, environmental purification strategies can help slow the spread of the virus. Unfortunately, current systems to treat circulated air are expensive and use primarily UV germicidal light. These products require professional installation, are not accessible to the general public per se, and have not been used to kill COVID-19. Moreover, filtration in an HVAC system can be ineffective. COVID-19 measures between 0.05 to 0.2 microns, but HEPA filters can filter particulate larger than 0.3 microns so additional protection is needed against the spread of COVID-19.

[0011] For these reasons, the subject matter of the present disclosure is directed to overcoming, or at least reducing the effects of, one or more of the problems set forth above.

SUMMARY OF THE DISCLOSURE

[0012] The subject matter of the present disclosure is directed to a purification device that filters air and seeks to destroy viruses, bacteria, mold, pollens, volatile organic compounds, allergens, and pollutants. The purification device is intended to be affordable, easily installed, accessible and useable in both residential and commercial settings. The purification device can be applied to real world solutions to best reduce viruses, such as COVID-19, and other pathogens in the circulating air, and the purification device can be deployed as a specialized heated filter for use in commercial, residential, mass transit, and public venues.

[0013] For example and as discussed below, the purification device includes a barrier heater or heated filter that uses targeted thermal conduction of high efficiency nickel foam/mesh raised to temperatures proven to kill pathogens, such as corona viruses (such as COVID-19). The purification device also includes an ultraviolet (UV) light source that uses UV-C light to destroy the virus. The UV light source and the barrier heater are combined together in a flame retardant and resistant filtration system, which can then be directly integrated into air returns, furnace intakes, and other parts of an air handling system of a facility or populated environ, such as an airport terminal, church, hospital, workshop, office space, residence, transit vehicle, school, hotel, cruise ship, recreational venue, etc. As there is no current cure for COVID-19 and many other pathogens, environmental purification strategies can help slow the spread of the virus, and the air purification provided by the disclosed device can provide a primary defense against transmission.

[0014] In one configuration, an apparatus is used with supplied power for treating air flow of an air

handing system of a facility. The apparatus comprises a frame, a filter, and an UV light source, and a heater. The frame has a plenum with an inlet and an outlet and is configured to position in the air flow of the air handing system for passage of the air flow therethrough.
[0015] The filter is disposed across a surface area of the plenum and comprises a first material, such as a metal. The filter is configured to filter the air flow therethrough up to a filtration threshold. The ultraviolet light source is disposed in the plenum. The ultraviolet light source is connected in electrical communication with the supplied power and is configured to generate an active field of ultraviolet radiation in the plenum. The heater is disposed across the surface area of the plenum and comprises a permeable barrier of metal material. The permeable barrier of the heater is configured to impede the air flow therethrough up to an impedance threshold. Moreover, the permeable barrier of the heater is connected in electrical communication to the supplied power and is heated to a surface temperature.
[0016] In another configuration, an apparatus is used with an air filter and supplied power for treating air flow of an air handing system in a facility. The apparatus comprises a frame, a UV light source, and a heater similar to that disclosed above. The filter can be mounted adjacent the frame or separately in the air handling system.
[0017] In yet another configuration, a method is used for treating air flow of an air handing system in a facility. A frame is positioned in the air handling system for passage of the air flow therethrough. The air flow is filtered up to a filtration threshold through a filter disposed across a surface area of a plenum of the frame between an inlet and outlet. An active field of ultraviolet radiation is produced in the plenum by powering an ultraviolet light source disposed in the plenum. The air flow is impeded up to an impedance threshold through a permeable barrier of heater disposed across the surface area the plenum and having a metal material. The permeable barrier of the heater is heated to a surface temperature by supplying a voltage potential across the permeable barrier.
[0018] The present disclosure may be directed to a purification device for treating air flow of an air handling system that comprises a plenum that comprises an inlet and an outlet, and the plenum can be configured for passage of the air flow into the inlet and out of the outlet thereof. A high-efficiency particulate air (HEPA) filter can be disposed across the plenum and can be configured to filter over 99 percent of particles of the air flow that have a particle diameter of at least 0.30 microns. A heater can be disposed across the plenum and may comprise a permeable barrier configured to allow the air flow to pass therethrough. A metal material of the permeable barrier can be connected in electrical communication to a supplied power. The permeable barrier can have an active surface area configured to interact with a pathogen of the air flow and can be heated by the supplied power to a surface temperature directed to kill the pathogen. The supplied power may not be electrically connected to the high-efficiency particulate air (HEPA) filter such that the high-efficiency particulate air (HEPA) filter may not be electrically connected to the supplied power and not heated while the heater is electrically connected to the supplied power and heated.
[0019] The present disclosure may also be directed to a method for treating air flow of an air handling system, that comprises filtering the air flow through a high-efficiency particulate air (HEPA) filter disposed across a plenum of the air handling system, the plenum being configured for passage of the air flow therethrough; after filtering the air flow through the high-efficiency particulate air (HEPA) filter, allowing the air flow to pass through a permeable barrier of a heater disposed across the plenum, the permeable barrier having a metal material and an active surface area, the active surface area being configured to interact with a pathogen of the air flow; and heating the active surface area of the permeable barrier of the heater to a surface temperature directed to kill the pathogen by supplying a voltage potential across the permeable barrier. No power may be applied to the filter such that the filter may not be electrically connected to the power and not heated while the heater is electrically connected to the power and heated.

[0020] The foregoing summary is not intended to summarize each potential embodiment or every aspect of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. **1** illustrates a facility having an air handling system having a purification device according to the present disclosure.

[0022] FIGS. **2**A, **2**B, **2**C, **2**D, and **2**E illustrate other arrangements of the disclosed

[0023] purification device used with various air handling systems.

[0024] FIGS. **3**A, **3**B, and **3**C illustrate front, side, and end views of a purification device of the present disclosure.

[0025] FIGS. **4**A and **4**B illustrate side schematic views of a purification device with an arrangement of its components.

[0026] FIG. **4**C illustrates a side schematic view of another purification device with an arrangement of its components.

[0027] FIGS. **5**A, **5**B, and **5**C illustrate graphs detailing features of a barrier heater for the disclosed purification device.

[0028] FIG. **6**A illustrates another heating arrangement having a plurality of electric elements disposed in a plenum of a frame and connected to a power source control.

[0029] FIGS. **6**B, **6**C, and **6**D illustrate other configurations of the disclosed purification device.

[0030] FIG. **7** illustrates a schematic arrangement of an air handling system having a number of purification devices.

[0031] FIG. **8**A illustrates a configuration having a number of purification devices subject to a master control unit.

[0032] FIG. **8**B illustrates another configuration having environmental components and several purification devices subject to a master environmental control.

[0033] FIGS. **9**A-**9**B illustrates side views of the permeable barrier for the disclosed heater in flat and corrugated configurations.

[0034] FIGS. **10**A-**10**B illustrate graphs for the barrier heater having a flat configuration.

[0035] FIGS. **11**A-**11**B illustrate graphs for the barrier heater having a corrugated configuration.

[0036] FIG. **12** illustrates a graph of exposure times and temperatures.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0037] The subject matter of the present disclosure is directed to a purification device for instantaneously eradicating pathogens, such as COVID-19 virus, from the circulating air by filtering and exposing the pathogens to high temperatures (above 200° C.) (above 392° F.). By doing so, the subject matter of the present disclosure can decrease infectious transmission of a virus and other biological species that may cause future pandemics, while providing a sense of security and peace of mind for the public to return to work, school, life, recreation and healthcare in a post-COVID-19 world.

[0038] The primary mechanism of action of the purification device is a specialized heated filter or barrier heater that uses a low energy, targeted thermal conduction of high performance, high resistant porous metal foam incased in a flame retardant frame. The disclosed heated filter or barrier heater can be combined with a highly-efficient HVAC filter. Additionally, ultraviolet light (UV-C) can be added to the system milieu for additive killing effect. Research has shown heat and low wavelength light have been proven to successfully deactivate COVID-19 with duration of exposure.

[0039] As disclosed below, the purification device of the present disclosure can be incorporated into air handlings systems of a facility, vehicle, or any other environment. Using the same technology, mobile/robotic COVID-19 purification device can be deployed for use in public venues, healthcare facilities, nursing homes, schools, airplanes, trains, cruise ships, performance venues, theaters, churches, grocery and retail stores, prisons, etc. Details are provided in related U.S. application Ser. Nos. 16/883,981 and 18/213,644, entitled “Mobile Purification Device Having Heated Filter for Killing Biological Species, Including COVID-19”, each of which is incorporated by reference.

[0040] As shown in FIG. **1**, a facility **10**, such as a home, hospital, office space, airport terminal, church, or other enclosed environment, has an air handling system **20**. As shown here, the system **20** is a heating, ventilating, and air conditioning (HVAC) system, although other air handling systems can be used. As is typical, the HVAC system **20** includes returns **30**, chases **32**, return ducts **34**, etc. that direct drawn return air from an indoor space to a blower **22**, heat exchanger **24**, and coiling coil **26** of the system **20**. In turn, the system **20** provides conditioned supply air to the space through supply ducts **36**, vents **38**, and the like. The heat exchanger **24** can include an electric or gas furnace for heating the air. The cooling coil **26** can be an evaporator connected in a cooling circuit to other conventional components outside the facility, such as a condenser, compressor, expansion valve, etc.

[0041] Integrated with or incorporated into this system **20**, one or more purification devices **100** are used in the facility to purify the air flow. In one arrangement and as shown, the purification device **100** is used in the air return **30** of the HVAC system **20**, through which return air is drawn to pass through the conditioning elements of the HVAC system **20**. Each air return **30** in a facility may have such a purification device **100** so return air is drawn through the purification device **100** during operation of the HVAC system **20**. Because HVAC systems **20** use a number of different filters of various sizes, the purification device **100** can have dimensions to suit various filter sizes.

[0042] As discussed in more detail later, the purification device **100** tends to heat the return air with flash heating. For this reason, the device **100** is preferably disposed in the return air upstream of the cooling unit **26**. This can allow some of the heat to be dissipated in the air flow before being cooled by the cooling unit **26**. When heating the indoor space, the purification device **100** may simply add to the heat provided by the system **20**. It is even conceivable that the vents **28** of the system **20** distributing air could have such purification devices **100**. However, the devices **100** may tend to diffuse the air flow and pushing air flow through filters is less efficient, making use of the devices **100** in vents possible but less favorable.

[0043] Study of airflow in a meeting room and office space shows that convection patterns can persistently carry infection between chairs at a conference table and between cubicles in an open office space. This shows that reliance on separation between people can be ineffective due to the convention of the air.

[0044] Control of the purification device **100** can be handled entirely by a local controller **200**, which determines independently if air flow is being conducted through the device **100**. Alternatively, the local controller **200** can be integrated with a system controller **50** for the HVAC system **20**, which can signal activation of the system **20** and indicate to the local controller **200** that air flow is being conducted through the device **100**. In a further alternative, the purification device **100** may lack local controls and may be centrally controlled by the system controller **50**. As will be appreciated, these control arrangements can be used in any combination throughout a facility **10**, multiple purifications devices **100**, conditioning zones, HVAC components, and the like.

[0045] Although FIG. **1** shows the purification device **100** disposed at the return **30** for the chase **32** of the air handling system **20**, other arrangements can be used. Overall, the purification device **100** can be sized for use in a typical furnace opening (14-20 in×25 in) as used commercially. Multiple HVAC zones can then be targeted by the purification device.

[0046] For example, FIG. **2**A shows a purification device **100** disposed immediately upstream of the blower **22** and other components of an HVAC system **20**, which has a horizontal furnace **24**. FIG. **2**B shows the purification device **100** disposed adjacent the blower **22** and other components of a system **20**, such as a horizontal furnace. Finally, FIG. **2**C shows the purification device **100** disposed above the blower in a downflow furnace. These and other configurations can be used. The furnace can use gas burners or electric heating elements as the case may be, and other conditioning components can be installed further downstream.

[0047] FIG. **2**D illustrates an air handling system **80** in an airplane **70** having a purification device **100** of the present disclosure. In the airplane **70**, the air in the cabin **74** may be changed 20 to 30 times per hour with about half of the air being recycled through filters. Because the cabin **74** is pressurized, outside air enters an inlet **82** of the system **80** at high temperature and pressure from the engines **72**. The hot and compressed air reaches the air conditioning units **84** for the plane **70** where the air is cooled considerably. For heating, some of the input air can enter the cabin **74** though the overhead outlets **75**. For cooling, the air from the conditioning unit **84** passes to a mixing manifold **86***a*, wherein the cooled outside air is combined with cabin air to produce a 50/50 mix. The mixed air from the mixing manifold **86***a* can then be circulated through the cabin **74** via the overhead outlets **75**. A portion of the air in the cabin **74** from inlets **77** is then discharged from outlet **79** in an equal amount to the outside air entering the cabin **74** to maintain a balance, and another portion of the cabin air though a buffer manifold **86***b* is recirculated in the mixing chamber **86***a*. Because the outside air is new, the purification device **100** of the present disclosure is placed at the mixing manifold **86***a* and/or the buffer manifold **86***b* of the air handling system **80** to treat the recycled cabin air.

[0048] FIG. **2**E illustrates an air handling system **90** used in a cruise ship having a purification device **100** of the present disclosure. As shown, return/relief air pulled through a return duct **92***a* is diverted through filters **94** by blowers **96***a*, which force the air through a heat wheel **98**. Additional blowers **96***b* then pass air out an exhaust **93***a* to the atmosphere.

[0049] Meanwhile, outside air entering intake **92***b* passes through filters **94**, and the other end of the heat wheel **98** before passing on to cooling and filtering elements. At the return duct **92***a*, the return/relief air is also diverted to the cooling and filtering elements. For these elements, the air is passed through filters **94**, cooling coils **95**, UV light treatment **97**, additional filters **94**, and a steam humidification treatment **99** before passing out to supply air ducts **93***b*.

[0050] As shown in FIG. **2**E, the purification device **100** can be used in the return air from the return duct **92***a* that is recycled back through the system **90**. Throughout the cruise ship, various components are used for conducting the air, including duct heaters, axial fans, dampers, etc. Various self-contained unit heaters can also be used in different areas of the cruise ship. Because the cruise ship is much like a facility, the purification device can be incorporated into the various returns, ducts, vents, and standalone units used throughout the vessel.

[0051] As will be appreciated, other vehicles and mass transit systems having air handling systems can benefit in a similar way to an airplane and a cruise ship. For example, busses, trains, and subways used in mass transit have air handling systems that typically use both outside air and recycled air. The disclosed purification device **100** can be incorporated into these air handling systems in a comparable way to those discussed above.

[0052] With an understanding of how the purification device **100** is used and where it can be installed in a facility, discussion now turns to particular details of the disclosed purification device **100**. FIGS. **3**A, **3**B, and **3**C illustrate front, side, and end views of an example purification device **100** of the present disclosure. The device **100** includes a frame **110** configured for insertion into an existing air return of a facility, for replacing an existing return altogether, or for use at the intake of a furnace.

[0053] Overall, the frame **110** has four sidewalls enclosing a plenum inside **116**, which is exposed on opposing open faces (one for an inlet **112** and another for an outlet **118** of the plenum **116**). If necessary, the inlet **112** can include a rim **114**, which would typically engage around a wall opening for a return (**30**: FIG. **1**). Fasteners (not shown) can affix the rim to surrounding structures. Although configured for a particular implementation, a typical size for the frame **110** may include overall dimensions of 20-in width×30-in height×7-in depth.

[0054] As best shown in FIG. **3**A, the inlet **112** or the rim **114** may form a receptacle for holding a filter (not shown) to filter entering air flow into the plenum **116**. Inside the plenum **116**, the frame **110** holds a barrier heater **140**. As briefly shown here, the barrier heater **140** includes a permeable barrier **142**, composed of metal and comprising a mesh, a foam, a screen, or a tortuous media, supported by a surrounding case **145** and disposed across the plenum **116** to provide a permeable surface area for treating the air flow as discussed below.

[0055] Also inside the plenum **116**, the frame can hold an UV light source **130** as an additional treatment in conjunction with the barrier heater **140**. (Other embodiments disclosed herein may not include the UV light source **130**.) As briefly shown here, the UV light source **130** includes two UV-C light emitting diode (LED) strips placed across the plenum **116** to provide an active field for treating the air flow as discussed below. More or fewer sources **130** can be used, and different types of sources **130** can be installed.

[0056] Turning to FIG. **4**A, a side schematic view of a purification device **100** is shown having an arrangement of its components. As noted previously, the purification device **100** can be used in a return **30** of an air handling system. The wall opening of the return **30** may typical have a return air grille **31** to protect internal components. The frame **110** of the purification device **100** fits in the return **30** and can be held by fixtures (not shown), such as bolts and screws. As noted, the air filter **120** can fit into a receptacle of the frame **110**. Typically, the filter **120** simply fits snuggly in the receptacle, but fastening could be used.

[0057] Preferably, the purification device **100** first filters the air flow up to a filtration threshold through the filter **120**. In this way, the filter **120** can keep out dust and other particulates from being drawn into purification device **100** and from being drawing further into the HVAC system (**20**: FIG. **1**).

[0058] As noted herein, an active field of ultraviolet radiation can be produced in the plenum **116** of the device **100** by powering the UV light source **130** disposed in the plenum **116**. In the plenum of the device **100**, the air flow is impeded up to an impedance threshold through the barrier heater **140** disposed in the plenum **116**. The barrier heater **140** includes a permeable barrier **142** (e.g., mesh, foam, screen, tortuous media) of a metal material, such as nickel, nickel alloy, titanium, steel alloy, or other metal material. The permeable barrier **142** can be flat, corrugated, bent, pleated, or the like and can be arranged in one or more layers. The metal mesh/foam **142** of the heater **140** is heated to a surface temperature by supplying a voltage potential across the mesh/foam. Preferably, the UV light source **130** is disposed in the plenum **116** between the filter **120** and the barrier heater **140** so that the irradiation from the source **130** can treat passing air flow and can also treat exposed surfaces of the filter **120** and the barrier heater **140**.

[0059] Turning now to FIG. **4**B, another side schematic view of a purification device **100** is shown having an arrangement of its components. The frame **110** of the device **100** is shown holding the filter **120**, UV light source **130**, and barrier heater **140** in the plenum **116**. The purification device **100** is used with control circuitry and supplied power. For example, the control circuitry includes a controller **200** having appropriate power circuitry and processing circuitry for powering and controlling the purification device **100**. The controller **200** can be connected to one or more types of power supply(s) **40**, such as available AC power supplies of a facility, battery power, or other power source. Power circuitry of the controller **200** can convert the supplied power as needed to produce DC power and voltage levels.

[0060] Looking at the frame **110**, the filter **120** is disposed in the plenum **116** of the frame **110** and can be held in a receptacle **115** toward the inlet **112**. The filter **120** is composed of a first material and is configured to filter the air flow therethrough up to a filtration threshold. Preferably, the filter **120** is a metal filter media **122** composed of stainless steel, aluminum, etc. that is meshed in one or more layers depending on the amount of air flow and the level of filtration required. The filter **120** has a case **125**, which is also composed of metal and which frames the metal filter media. In general, the metal filter **120** can be a 1-in thick HVAC filter made from metal that is fire resistant and retardant and that has a high efficiency rating.

[0061] The barrier heater **140** is also disposed in the plenum **116** and can be situated toward the outlet **118**. Insulation **145** for both heat and electricity may separate the barrier heater **140** from the frame **110**. The barrier heater **140** includes a mesh/foam of a metal material and is configured to impede the air flow therethrough up to an impedance threshold.

[0062] The UV light source **130** can be disposed in the plenum **116** and, as noted previously, can preferably be situated between the metal filter **120** and barrier heater **140**. The UV light source **130** produces an active field of UV-C light in the plenum **116** to treat passing air flow. As noted herein, pathogens, such as viruses, can be eliminated when subjected to a dose of ultraviolet light. For example, the sRNA coronavirus up to 0.11 μm in size can be eliminated >99% with only about 611 μj/cm2 UVGI dose.

[0063] Both the UV light source **130** and the barrier heater **140** are connected in electrical communication with the power supply **40** through the controller **200**, which controls the illumination of the light source **130** and the heating of the barrier heater **140** in the plenum **116**.

[0064] The UV light source **130** can include one or more UV-C lamps, a plurality of light emitting diodes, or the like disposed in the plenum **116**. For example, the source **130** can use one or more Ultraviolet Germicidal lamps, such as mercury-vapor lamps. The source **130** can also use light emitting diodes having semiconductors to emit UV-C radiation.

[0065] One or more structures can be disposed in the frame **110** to support the UV light source **140**. The structures used can depend on the type of source **140** used and can include fixtures for lamps and strips for UV-C LEDs. For example, the UV light source **130** can uses several strips of UV-C light emitting diodes stretched across the plenum **116**.

[0066] The effectiveness of the UVGI treatment in the air flow depends on a number of factors, including the targeted microbial species, the intensity of the exposure, the time of the exposure, and the amount of humidity in the air. A sufficient dose will kill the DNA-based microorganism. Therefore, the intensity of the UVGI treatment, the time for exposure, and other factors can be configured and further controlled in the purification device **100** and HVAC system to reach a desired effectiveness.

[0067] The UVGI treatment provided by the purification device **100** can be effective at destroying pathogens, such as COVID-19. The UV-C or short-wave light generated by the UV light source at wavelengths from 100-280 nanometers may have a proven germicidal effect. In particular, 222 nanometer low, far-UVC light is effective at killing and deactivating aerosolized virus with duration of exposure.

[0068] In contrast to conventional use of UVGI in an HVAC system, the disclosed purification device **100** does not require high costs and special installation in air returns or duct systems. Rather, the disclosed device **100** offers practical installation and operation that can be seen as easy as changing an HVAC filter every 1-3 months in your home.

[0069] As discussed in more detail below, the metal permeable barrier of the barrier heater **140** can include nickel mesh/foam. The barrier heater **140** is configured to impede the air flow therethrough up to the impedance threshold of 20 percent, giving the foam a porosity of at least 80%.

[0070] The purification device **100** can include anti-microbial coatings on one or more surfaces to eliminate live bacteria and viruses. For example, the filter **120** can have an anti-microbial coating to eliminate pathogens trapped by the filter media. The inside walls of the frame's plenum **116** can also have anti-microbial coating. If practical under heated conditions, the mesh/foam of the barrier heater **140** can have anti-microbial coating.

[0071] As further shown in FIG. **4**B, the controller **200** disposed in electrical communication with the UV light source **130** and the barrier heater **140** is configured to control (i) the radiation of the UV light source **130** powered by the power supply **40**, and (ii) the heating of the barrier heater **140** by the power supply **40**. This controller **200** can be a local controller that can include a communication interface **212** to communicate with other purification devices and with other components of an air handing system (**20**: FIG. **1**) in a facility, such as a system controller (**50**). The local controller **200** can receive a signal that the HVAC system (**20**) is on/off, which is indicative of the passage of the air flow through the device **100**. The controller **200** can then the control the heating of the barrier heater **140** and the illumination of the UV light source **130** based on the signals received.

[0072] To do this, the controller **200** is disposed in electrical communication with heater circuitry **214** connected to the barrier heater **140**. At least for a period of time when air is passed through the device **100** (being drawn by the HVAC system), the controller **200** can control the heating of the barrier heater **140** with the heater circuitry **214** powered by the power supply **40**. As will be appreciated, the controller **200** and heater circuitry **214** includes any necessary switches, relays, timers, power transformers, etc. to condition and control power supplied to the barrier heater **140**.

[0073] The controller **200** heats the barrier heater **140** at least while the controller **200** is signaled that the HVAC system (**20**) is operating indicating air flow through the device **100**. Pre-heating before the HVAC system (**20**) draws return air can occur before air is drawn through the device **100** so that a target temperature can be reached beforehand. This may require an advance signal from the system controller (**50**) or may involve intermittent heating of the barrier heater **140** to maintain some base temperature. Post-heating after the HVAC system (**20**) turns off may also be beneficial for a number of reasons.

[0074] The controller **200** is also disposed in electrical communication with drive circuitry **213** connected to the UV light source **130**. At least for a period of time when air is passed through the device **100** (being drawn by the HVAC system), the controller **200** can control the illumination of the UV light source **130** with the drive circuitry **213** powered by the power supply **40**. As will be appreciated, the controller **200** and drive circuitry **213** includes any necessary switches, relays, timers, power transformers, electronic ballast, etc. to condition and control power supplied to the light source **130**.

[0075] At least when the controller **200** is signaled that the HVAC system (**20**) is operating indicating air flow through the device **100**, the controller **200** illuminates the light source **130**. To reach a target illumination, some pre-lighting may be necessary for the lamps or the like of the UV light source **130** to reach full illumination before air is drawn through the device **100**. This may require an advance signal from the system controller (**50**). Post-lighting of the source **130** after the HVAC system (**20**) turns off may also be beneficial for a number of reasons.

[0076] For monitoring and control, the controller **200** can include one or more sensors **216**, **217**, and **218**. For example, the controller **200** can include a temperature sensor **216** disposed in the plenum **116** adjacent the barrier heater **140** and disposed in electrical communication with the controller **200**. The temperature sensor **216** is configured to measure temperature associated with the heating of the barrier heater **140** so the controller **200** can reach a target temperature. Depending on the implementation and the pathogens to be affected, the barrier heater **140** can heated to the surface temperature over about 54° C. (130° F.). In fact, research shows that heat at about 56° C. or above 56-67° C. (133-152° F.) can kill the SARS coronavirus and that 222-nanometer far-UVC light can be effective at killing and deactivating aerosolized virus.

[0077] The controller **200** can be connected to a light sensor **28**, such as a photocell or other light sensing element, to monitor the illumination, intensity, wavelength, operation, and the like of the UV light source **130**. For example, the UV light source **130** can be configured to produce the ultraviolet radiation with at least 611 μJ/cm2 dose of ultraviolet germicidal irradiation in an active field in the plenum **116**, and measurements from the light sensor **218** can monitor the radiation.

[0078] The controller **200** can be connected to yet another sensor **217**, such as a flow sensor to sense flow, velocity, and the like of the air passing through the plenum **116**. The detected flow by the flow sensor **217** may be used by the controller **200** to initiate operation of the device **100** if not signaled remotely. The velocity of the flow may be measured by the flow sensor **217** to coordinate a target flow velocity through the device **100** so the heating of the air flow by the barrier heater **140** can be coordinated to the detected flow velocity and a target heating level. Should the device **100** be integrated with the HVAC system (**20**) operable at different flow levels, then feedback from the flow sensor **217** can be used to control or indicated the level of drawn air through the device **100**. The velocity of the flow may also be monitored to coordinate the target irradiation of the air flow by the UV light source **140** so appropriate exposure levels can be achieved.

[0079] As noted herein, the purification device **100** combines thermal energy with UV-C light and is constructed within a flame retardant and resistant filtration system. The device **100** can be placed in a return behind the HVAC grill for return air. As disclosed herein, embodiments of the purification device **100** include the barrier heater **140** and can thereby include the various features of the controller **200**, sensors, etc. discussed above for the barrier heater **140**. Some embodiments may not include the UV light source **130**, while other embodiments may include the UV light source **130** along with the various features of the controller **200**, sensors, etc. discussed above for the UV light source **130**. In particular, FIG. **4**C illustrates another side schematic view of a purification device **100** having an arrangement of its components without a UV light source. Similar components are provided the same reference numerals as other embodiments and are not repeated here.

[0080] As proposed, the disclosed purification device **100** can eliminate pathogens, such as COVID-19, while filtering the air to 99.97% (ASME, U.S. DOE) of particles. As disclosed in the co-pending application incorporated herein, the configuration can be combined into a mobile housing for use at larger public venues to include airport terminals, churches, hospitals and other enclosed areas to reduce infectious air particles.

[0081] Although the purification device **100** has been described above as including a frame **110** that accommodates an air filter in the frame **110**. The device **100** can include a frame **110** that mounts behind a conventional air return **30** already accommodating a filter. Alternatively, the device **100** can include a frame **110** that mounts at an intake of a furnace downstream from a separately held air filter **120**. The purification device **100** can be sized to a furnace opening (e.g., 14-20 in×25 in) for commercial use. HVAC zones can then be targeted. In this types of arrangement, the purification device **100** can include a frame **110**, an UV light source **130**, and a barrier heater **140** as before, but the frame **110** does not necessarily hold or receive an air filter **120**. Instead, separate air filters can be installed elsewhere in the HVAC system, such as at returns.

[0082] Discussion now turns to details of the barrier heater **140** of the disclosed purification device **100**. The metal mesh/foam of the barrier heater **140** can have one or more layers of material and can have a suitable thickness. As one example, the mesh/foam may have thickness of 0.5 mm to 2.0 mm. Composed of nickel (Ni), the metal mesh/foam may have surface charge density (G) of 1.43×107C/m2. The Ni mesh/foam is electrically conductive, and it is highly-porous having random three-dimensional channels defined therethrough. The mesh/foam exhibits resistance of about 00.178Ω, and the electrical resistivity of an exemplary Ni foam is calculated to be about 1.51×10−5 Ωm.

[0083] For example, FIG. **5**A illustrates a first graph **60**A of the temperature (° C.) produced by an exemplary Ni foam material for the barrier heater per unit of supplied power (W). A sample of the foam having a size of 1.65 mm×195 mm×10 mm was investigated. The temperature was measured after applying voltage until the temperature became stable. As shown in the graph **60**A, the temperature is shown to rise generally linearly per unit supplied power so that about 7 watts produces a temperature of about 120° C. (248° F.).

[0084] FIG. **5**B illustrates a second graph **60**B of the measured temperature of gas (e.g., N2) after flowing through the exemplary Ni foam material for the barrier heater heated to a temperature. The gas for the measurement originated from an upstream distance of about 3.5 cm from the heated Ni foam material while room temperatures was about 21.7°° C. (71° F.). Temperature measurements were made at different downstream distances relative to the exemplary Ni foam material, which was heated to an initial temperature of about 115° C. (239° F.). As can be seen, the measured temperature of the gas decreased from about 29° C. to 23° C. (84° F. to 73° F.) for downstream distances ranging from 1 cm to 4 cm from the exemplary Ni foam material. This indicates that the heating produced by the barrier heater **140** composed of such an exemplary Ni foam material provides a tortuous heated surface area against which the air flow and any pathogens can impact, but the heating is localized and dissipated in the downstream air flow.

[0085] FIG. **5**C illustrates another graph **60**C of the measured temperature made at different downstream distances relative to the exemplary Ni foam material at another initial temperature. Here, the Ni foam material is at an initial temperatures of about 54° C. (129° F.). The measured temperature of the gas decreased from about 24.5° C. to 21.7° C. (76° F. to 71° F.) for distances ranging from 1 cm to 4 cm from the exemplary Ni foam material.

[0086] As noted herein, the barrier heater **140** can use nickel, but could use nickel-based alloys or iron-based alloys developed for applications at high service temperatures and in corrosive environments. Nickel is slowly oxidized by air at room temperature and is considered corrosion-resistant. Nickel is a high performance metal that can be easily regulated to reach high temperatures with minimal transmission of heat to its surroundings or to air molecules passing through it. When voltage is passed through nickel mesh/foam (1.43×107δ), for example, the metal conducts energy to a target temperature hot enough to kill pathogens, including COVID-**19** on contact. The target temperature can be (56° C. to 66° C. or more, and even over 93° C.) (133° F. to 150° F. or more, and even over 200° F.). In this way, the Ni mesh/foam (0.5 mm-2.0 mm) provides a heated, charged surface area for the pathogen to impact and be eliminated by the heated latticework. Meanwhile, the porosity (80-90%) of the foam/mesh of the barrier heater **140** does not overly impede the air flow and does not detrimentally increase the energy required from the HVAC system.

[0087] As disclosed above, heating in the plenum **116** can be achieved with the barrier heater **140** having the mesh/foam that is heated to the target temperature and provides a tortuous path for return air passing through the mesh/foam. Other forms of heating can be used. As disclosed above, UV illumination in the plenum **116** can be achieved with UV light strips. Other forms of UV illumination can be used.

[0088] For example, FIG. **6**A shows another arrangement having a plurality of electric elements (UV light sources **130** and a barrier heater **140**) disposed in a plenum **116** of a frame **110** and connected to a power source control **201**. The plenum **116** includes carbon media **152** on one or more sidewalls for absorption and purification purposes. The plenum **116** may also include a filter **120** that is disposed at the inlet.

[0089] As hinted to above, the disclosed purification device **100** can be used separately or in combination with an air handling system and other purification devices **100**. As one example, FIG. **6**B shows a configuration of a purification device **100** according to the present disclosure, which includes a UV light source **130** and a barrier heater **140** controlled by control/power circuitry **202**. The UV light source **130** and the barrier heater **140** can be similar to those disclosed herein and can be housed together in a housing or frame **110** to fit into the air flow of an air handling system. For example, the housing or frame **110** can be retrofitted or added to an existing duct of the air handling system, can be disposed upstream of operable components of the air handling system, or can be configured elsewhere in the air flow. Filtering can be achieved elsewhere in the air handling system. For its part, the control/power circuitry **201** may have the necessary components as disclosed herein to control the UV light source **130** and the barrier heater **140**.

[0090] As another example, FIG. **6**C shows another configuration of a purification device **100** according to the present disclosure, which includes a barrier heater **140** controlled by control/power circuitry **203**. This device **100** as shown may not include a UV light source, although such a source could be used elsewhere in a facility or other environment. The barrier heater **140** can be similar to those disclosed herein and can be housed in a housing or frame **110** to fit into the air flow of an air handling system. For example, the housing or frame **110** can be retrofitted or added to an existing duct of the air handling system, can be disposed upstream of operable components of the air handling system, or can be configured elsewhere in the air flow. Filtering can be achieved elsewhere in the air handling system or can be incorporated into the frame **110** using a filter (not shown) as disclosed elsewhere herein. For its part, the control/power circuitry **203** may have the necessary components as disclosed herein to control the barrier heater **140**.

[0091] As yet another example, FIG. **6**D shows yet another configuration of a purification device **100** according to the present disclosure, which includes a UV light source **130** controlled by control/power circuitry **204** and includes a barrier heater **140** controlled by control/power circuitry **203**. The UV light source **130** and the barrier heater **140** can be similar to those disclosed herein and can be housed in separate housings or frames **110***a-b* to fit into the air flow of an air handling system. For example, the housings or frames **110***a-b* can be retrofitted or added to existing ducts of the air handling system, can be disposed upstream of operable components of the air handling system, or can be configured elsewhere in the air flow. Filtering can be achieved elsewhere in the air handling system or can be incorporated into either one or both of the frames **110***a-b* using a filter (not shown) as disclosed elsewhere herein. For their parts, the control/power circuitry **203**, **204** may have the necessary components as disclosed herein to control the UV light source **130** and the barrier heater **140** respectively.

[0092] As hinted to above, the disclosed purification device **100** can be used separately or in combination with an air handling system and other purification devices **100**. FIG. **7** illustrates a schematic arrangement of an air handling system **20** having several purification devices **100***a-n*. As noted above, more than one purification device **100***a-n* can be used in a facility, and these devices **100***a-n* can have control configurations for remote or localized control.

[0093] For example, the air handling system **20** (e.g., HVAC system) can include its system controller **50** and can have user/communication interfaces **52**. The system controller **50** includes a central processing unit and memory as typically found in environmental controllers. The user/communication interfaces **152** can include graphical user interfaces, control panels, wired communications, and wireless communications, such as typically found in environmental controllers. As before, the HVAC system **20** includes components, such as a blower **22**, a furnace **24**, a compressor **27**, thermostats **29**, and any other convention components.

[0094] The system controller **50** can communicate via wired or wireless communication with one or more stand-alone purification devices **110***a*-**100***n* arranged in the facility. These stand-alone purification devices **110***a*-**100***n* have local controllers **210** and user/communication interfaces **212**. The local controller **210** includes a central processing unit and memory as typically found in environmental controllers. The user/communication interfaces **212** can include graphical user interfaces, control panels, wired communications, and wireless communications, such as typically found in environmental controllers. As before, the stand-alone devices **100***a*-**100***n* include the disclosed purification components, such as the UV source driver **213**, the heater circuitry **214**, the sensors **216**, etc.

[0095] As further shown, the system controller **50** can likewise communicate via wired or wireless communication with one or more integrated purification devices **110***b* arranged in the facility. These integrated devices **110***b* do not have local control and may be controlled directly by the system controller **50**. As before, the integrated device **100***b* includes the disclosed purification components, such as the UV source driver **213**, the heater circuitry **214**, the sensors **216**, etc.

[0096] Based on the above arrangement, it will be appreciated that the facility can be configured with multiple system components for different zones, rooms, areas, etc. of the facility. Briefly, FIG. **8**A shows a master control unit **250** having a central processing unit **252** and communication interfaces **245** for communicating via wired and/or wireless communications **256** with multiple local controllers **200***a-n* in different zones **104***a-n* of a facility configuration **102**. Each of the local controllers **200***a-n* can control one or more of the purification devices **100***a-n* in a given zone **104***a-n*.

[0097] As another brief example, FIG. **8**B shows a master environmental control **50** having a central processing unit and communication interfaces **52***a-b* for communicating via wired and/or wireless communications **56** with multiple system components in the facility configuration **102**. The master environment control **50** can communicate with local controllers **200***a-n* in different zones **104***a-n* of the facility configuration **102**. Each of the local controllers **200***a-n* can control one or more of the purification devices **100***a-n* in a given zone **104***a-n*. Additionally, the master control **50** can communicate with local environmental systems **21***a-n* of the facility's air handling system **20**. These local environmental systems **21***a-n* can be dedicated to different zones (e.g., floors, rooms, buildings, etc.) of a facility.

[0098] As noted previously, the permeable barrier **142** of the barrier heater **140** disclosed herein can have different layers and configurations. In FIG. **9**A, a portion of a barrier heater **140***a* is shown with the permeable barrier **142** being flat and having a defined thickness T**1**. One or more such flat barriers **142** can be used adjacent one another in series to impede and interact with the impinging air flow. To increase surface area and the interaction, a portion of a barrier heater **140***b* is shown in FIG. **9**B having folds, corrugations, or pleats **142** in the permeable barrier **142**. The mesh material of the barrier **142** may have its original thickness T**1**, but the corrugated barrier heater **140***b* presents a thickness of T**2** for the impinging air flow. One or more such corrugated barriers **142** can be used adjacent one another in series to impede and interact with the impinging air flow.

[0099] Considering the flexibility of Ni foam, the corrugated barrier heater **140***b* offers several advantages. First, the resistance of the Ni foam is much larger with the bends **144**, which can help the barrier heater **140***b* when used with the residential voltage (110 V). Second, as illustrated in FIG. **9**B, the bends **144** produce an effective distance T**2** multiple times greater than the thickness T**1** for interacting with the impinging air. The gaps between the bends **144** in the hot Ni foam create a high temperature that can be effective in damaging pathogens. It should be noted that the number of bends, the bending length, and the like can be easily controlled, and the longer the bending length, the higher the temperature can be achieved. Third, compared to the flat Ni foam with two main sides exposing to air, the bended Ni foam barrier **140***b* in FIG. **9**B has a much smaller area exposed to the incoming and outgoing air, which will minimize the heat loss, so temperature of the barrier heater **140** can increase more rapidly and can reach to a much higher value at the same power consumption.

[0100] For example, FIG. **10**A illustrates a graph of input voltage relative to the current produced for the barrier heater **140***a* having a flat Ni foam configuration, and FIG. **10**B illustrates another graph of the current relative to a temperature level produced for the barrier heater **140***a* having the flat Ni foam configuration. Meanwhile, FIG. **11**A illustrates a graph of input voltage relative to the current produced for the barrier heater **140***b* having corrugated Ni foam configuration, and FIG. **11**B illustrates another graph of the current relative to a temperature level produced for the barrier heater **140***b* having the corrugated Ni foam configuration. As can be seen in FIGS. **10**B and **11**B, under the same voltage of 1.0 V, the temperature of the corrugated barrier heater **140***b* can be more than twice that of the flat barrier heater **140***a*.

[0101] As will be appreciated, various features of the disclosed purification device **100** with its UV light source **130** and barrier heater **140** can be configured to meet a particular implementation and to treat air for particular pathogens. Testing with actual pathogens requires careful controls, which has been conducted in laboratory environments.

[0102] As to the UV light source **130**, the intensity, the active area, the wavelength, and other variables of the UV light from the source **130** can be configured to treat air for particular pathogens, and the variables are best determined by direct testing with actual pathogens in a controlled laboratory setting.

[0103] As to the barrier heater **140**, the thickness, material, active surface area, permeability, corrugations, temperature, and other variables of the permeable barrier **142** from the barrier heater **140** can be configured to treat air for particular pathogens, and the variables are best determined by

direct testing with actual pathogens in a controlled laboratory setting.
[0104] Previous studies with SARS-COV and MERS-COV have established that coronaviruses can be inactivated by heat. See e.g., See e.g., Leclerca, 2014; Darnell, 2004; Pastorino, 2020. Results of a preliminary study in a BSL3 facility showed SARS-COV-2 is remarkably heat-resistant for an enveloped RNA virus. Only the 100° C. (212° F.) for 10 minutes protocol totally inactivated the virus.
[0105] In particular, heat resistance of the Human SARS-COV-2 strain (COVID-19) has been conducted in a BSL3 facility. The protocols for the study included using water and saline either at room temperature or at boiling temperature (FIG. **12**). For the latter, 10 □L of SARS-CoV-2 was added to 90 □L of preheated water or saline at 100° C. (212° F.). These solutions were either incubated at 100° C. for either 30 seconds or 10 minutes whereas for the control arm incubation was carried out at room temperature.
[0106] After the incubation, 900 □L room-temperature media was added and titrated. The control arm of 10 min and 30 seconds incubation at room temperature remained ineffective in reducing the virus load. By contrast, the protocol 100° C.-30 seconds depicted a trend, but the exposure apparently was not long enough to effectively reduce the virus load, although the virus load in water was relatively lower compared to saline. Only the 100° C.-10 minute protocol for either water or saline was able to totally inactivate the virus (>5 Log10 decrease).
[0107] The generated data confirms the virus to be remarkably heat-resistant for an enveloped RNA virus. Additional studies about the heat inactivation can illustrate curves for variable temperatures (50° C., 100° C., 150° C., 200° C., 250° C. & 300° C.) and exposure durations (1 sec, 5 sec, 15 sec, 30 sec, 1 min, 3 min, & 5 min), which can then be correlated to the expected heat damage caused by a barrier heater as disclosed herein, such as having a permeable Ni foam.
[0108] According to recent research, however, the heated filter of the disclosed barrier heater **140** can be used safely at high temperatures [(200-250° C.) (392-482° F.)] to kill COVID-19. In particular, research has been conducted at the Galveston National Lab/NIAID Biodefense Laboratory Network (Biosafety Level 4) and include findings of controlled experiments. The research has found COVID-19 to be vaporized in aerosolized air on contact with the specialized heated filter system of the present disclosure (i.e., the disclosed barrier heater **140**). The results show a 100-fold decrease in active virus and a 100-percent kill rate of COVID-19 by the heated barrier heater **140**. This research shows how COVID-19 can be eliminated from the air.
[0109] The disclosed purification device **100** can kill viruses and bacteria in the cycling air efficiently at high temperatures of about 250° C. (482° F.). As disclosed herein, the barrier heater **140**, such as the nickel (Ni) foam, is low cost, electrically conductive, highly porous with random channels, and mechanically strong with good flexibility, which act as a good filter for sterilization and disinfection in an HVAC system or other environment. A bended Ni foam provides a structure with higher resistance and lower voltage and increases surface area for sterilization. A mechanical kill using temperature and a supercharged, high performance metal may be applied to the setting of COVID-19.
[0110] Other related research as disclosed herein has found that there is not a significant temperature increase in the air that passes through the disclosed heated filter given its high performance and design. Primary research of the filter and is conductivity was completed at the Superconductivity Center of Texas at The University of Houston. Research partners include Texas A&M University, Department of Engineering and Engineering Experiment Station and the University of Texas Medical Branch. As has been illustrated, the temperature of the barrier heater **140** of Ni foam increases very fast and can be heated to a high temperature with low wattage power. The air temperature decreases very fast after passing through the heated Ni foam of the barrier heater **140**, even at temperatures over 100° C. (212° F.), the air temperature is room temperature at 4 cm away.
[0111] The foregoing description of preferred and other embodiments is not intended to limit or

restrict the scope or applicability of the inventive concepts conceived of by the Applicants. It will be appreciated with the benefit of the present disclosure that features described above in accordance with any embodiment or aspect of the disclosed subject matter can be utilized, either alone or in combination, with any other described feature, in any other embodiment or aspect of the disclosed subject matter.

[0112] In exchange for disclosing the inventive concepts contained herein, the Applicants desire all patent rights afforded by the appended claims. Therefore, it is intended that the appended claims include all modifications and alterations to the full extent that they come within the scope of the following claims or the equivalents thereof.

Claims

1. A purification device for treating air flow of an air handling system, the purification device comprising: a plenum comprising an inlet and an outlet, the plenum being configured for passage of the air flow into the inlet and out of the outlet thereof; a high-efficiency particulate air (HEPA) filter disposed across the plenum, the high-efficiency particulate air (HEPA) filter being configured to filter over 99 percent of particles of the air flow that have a particle diameter of at least 0.30 microns; and a heater disposed across the plenum, the heater comprising a permeable barrier being configured to allow the air flow to pass therethrough, a metal material of the permeable barrier being connected in electrical communication to a supplied power, the permeable barrier having an active surface area being configured to interact with a pathogen of the air flow and being heated by the supplied power to a surface temperature directed to kill the pathogen, wherein the supplied power is not electrically connected to the high-efficiency particulate air (HEPA) filter such that the high-efficiency particulate air (HEPA) filter is not electrically connected to the supplied power and not heated while the heater is electrically connected to the supplied power and heated.

2. The purification device of claim 1, wherein the heater is located in the plenum downstream of the high-efficiency particulate air (HEPA) filter.

3. The purification device of claim 1, wherein the permeable barrier of the heater comprises a mesh or a foam.

4. The purification device of claim 1, wherein the permeable barrier comprises one or more layers of porous foam having a thickness, the active surface area comprising a latticework of tortuous channels defined through the thickness of the porous foam.

5. The purification device of claim 4, wherein the one or more layers are corrugated.

6. The purification device of claim 1, wherein the high-efficiency particulate air (HEPA) filter comprises a metal material that is different than the metal material of the permeable barrier.

7. The purification device of claim 6, wherein the metal material of the permeable barrier comprises nickel, nickel-based alloy, iron-based alloy, titanium, or steel alloy.

8. The purification device of claim 7, wherein the metal material of the high- efficiency particulate air (HEPA) filter is aluminum or stainless steel.

9. The purification device of claim 1, further comprising electrical insulation separating the permeable barrier from a frame of the plenum.

10. The purification device of claim 1, wherein the high-efficiency particulate air (HEPA) filter is disposed in the plenum toward the inlet thereof, and the heater is disposed in the plenum toward the outlet thereof.

11. The purification device of claim 1, further comprising a controller disposed in electrical communication with the permeable barrier and being configured to control the heating of the permeable barrier by the supplied power.

12. The purification device of claim 11, wherein the controller is configured to control heating of the active surface area of the permeable barrier intermittently, when the air flow is passing through the permeable barrier, and/or when the air flow is not passing through the permeable barrier.

13. The purification device of claim 11, wherein the controller is disposed in electrical communication with heater circuitry connected to the permeable barrier, and the controller is configured to control the heating of the permeable barrier with the heater circuitry powered by the supplied power.

14. The purification device of claim 11, further comprising a temperature sensor disposed adjacent the permeable barrier and disposed in electrical communication with the controller, the temperature sensor being configured to measure a temperature associated with the heating of the permeable barrier.

15. The purification device of claim 11, the controller comprises a communication interface disposed in communication with an air handling system and being configured to receive a signal indicative of the passage of the air flow through the purification device, and the controller configuring control based on the signal received.

16. The purification device of claim 11, further comprising a flow sensor disposed adjacent the plenum and disposed in electrical communication with the controller, the flow sensor being configured to measure the air flow passing through the plenum, and the controller configuring the control based on the measured air flow.

17. The purification device of claim 1, wherein the plenum is configured to be positioned in at least one of: a return of the air handling system in a facility; an intake of a furnace of the air handling system in a facility; an outlet of the air handling system in a facility; and a mixing chamber of the air handling system of a vehicle.

18. The purification device of claim 1, wherein the pathogen being a virus, wherein the active surface area of the permeable barrier is heated to the surface temperature of at least 200° C. directed to kill the virus.

19. A method for treating air flow of an air handling system, the method comprising: filtering the air flow through a high-efficiency particulate air (HEPA) filter disposed across a plenum of the air handling system, the plenum being configured for passage of the air flow therethrough; after filtering the air flow through the high-efficiency particulate air (HEPA) filter, allowing the air flow to pass through a permeable barrier of a heater disposed across the plenum, the permeable barrier having a metal material and an active surface area, the active surface area being configured to interact with a pathogen of the air flow; and heating the active surface area of the permeable barrier of the heater to a surface temperature directed to kill the pathogen by supplying a voltage potential across the permeable barrier, wherein no power is applied to the filter such that the filter is not electrically connected to the power and not heated while the heater is electrically connected to the power and heated.

20. The method of claim 19, the active surface area heated to the surface temperature and a porosity threshold of the permeable barrier are configured to produce heating that is localized and is dissipated in the air flow downstream thereof.